

Centrifugal Pump Operations & Maintenance Manual

Applies to: P-101, P-102, P-201 | Doc ID: OEM-PUMP-CM-001 | Rev: 5 | Date: 2024-01-10

1. Equipment Overview

P-101, P-102, and P-201 are horizontal end-suction centrifugal pumps rated at 45 m³/h at 32 m head. Operating speed: 1450 RPM driven by 11 kW induction motors. Mechanical seal type: single cartridge spring-loaded, API Plan 11 flush. Bearing arrangement: drive end 6309-2RS, non-drive end 6206-2RS.

2. Startup Procedure

Refer to SOP-PUMP-07 for the full startup sequence. Key pre-start checks: verify suction valve fully open, confirm seal flush pressure above 0.8 bar, check oil level at bearing housing sight glass, and ensure coupling guard is fitted.

3. Maintenance Schedule

Weekly: check bearing temperature (max 75 deg C), inspect seal flush flow (min 2 L/min), check for leakage at gland.

Monthly: verify coupling alignment (max 0.05 mm TIR), measure vibration.

6-Monthly: replace bearing grease (Mobilux EP2, 15 g per bearing).

Annual: replace mechanical seal cartridge MS-45-CR, inspect impeller for cavitation damage, replace wear ring if clearance exceeds 0.5 mm.

4. Vibration Standards (OISD-137)

Per OISD-137, maximum allowable vibration velocity is 4.5 mm/s RMS. Alarm setpoint: 3.5 mm/s. Trip setpoint: 7.0 mm/s. Any reading above 4.5 mm/s requires investigation within 48 hours. Measurements must be taken at the bearing housing in vertical, horizontal, and axial directions.

5. Common Failure Modes

1. Mechanical seal leakage: caused by misalignment greater than 0.1 mm TIR, incorrect flush pressure, or seal face damage from dry running. Corrective action: replace cartridge seal MS-45-CR and verify alignment per SOP-PUMP-07.

2. Bearing overheating above 75 deg C: caused by over-greasing, under-greasing, or contaminated lubricant. Replace bearing 6309-2RS if temperature exceeds 85 deg C.

3. High vibration above 4.5 mm/s: caused by misalignment, cavitation, worn impeller, or unbalanced rotor. Check shaft alignment first, then verify NPSH available.

4. Cavitation: occurs when suction pressure drops below NPSH_r of 3.2 m. Check suction strainer differential pressure and verify vessel level.

5. Motor overload trip: caused by high-density product, partially closed discharge valve, or impeller damage. Verify motor FLC on nameplate and check VFD parameters.

6. Spare Parts List

Mechanical seal cartridge: MS-45-CR (annual consumable)

Drive end bearing: 6309-2RS

Non-drive end bearing: 6206-2RS

Wear ring set: WR-P101-CR

Coupling insert: CI-HRC-70

Gland packing backup: GP-12.7-AF